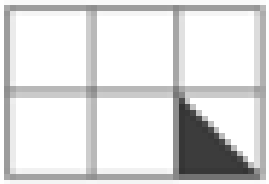
3 Points Each

2001

1. In the year 2001 the Little Kangaroo calculated that the value of the expression 2 x 0 + 0 × 1 is

(A) 2
(B) 0
(C) 1
(D) 2001
(E) 3

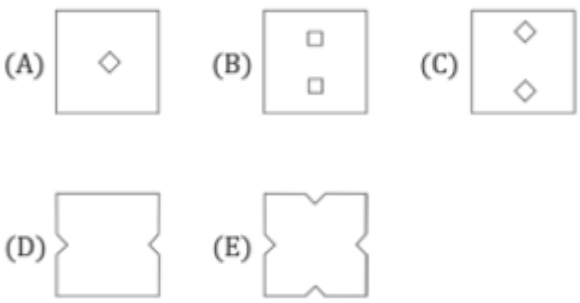
2. What part of the rectangle does the shaded region represent? (See the picture.)



A) $\frac{1}{6}$
(B) $\frac{1}{8}$
(C) $\frac{1}{10}$
(D) $\frac{1}{12}$
(E) $\frac{1}{15}$

3. Which of the 5 napkins below was made from the cut-out you can see on the right?





4. At noon grandpa set his clock to the correct time. The clock is running 20 seconds late per hour. After 24 hours, the clock will be late:

(A) 7 minutes
(B) 8 minutes
(C) 9 minutes
(D) 10 minutes
(E) 11 minutes

5. A certain plane can carry 108 passengers. During one of its flights, Anita noticed that not all the seats were occupied. There were twice as many occupied seats as empty ones. How many passengers were on the plane?

(A) 36
(B) 42
(C) 56
(D) 64
(E) 72

6. Johnny has 3 sisters and 5 brothers. His sister Ella found that the product of the number of her sisters and the number of her brothers is:

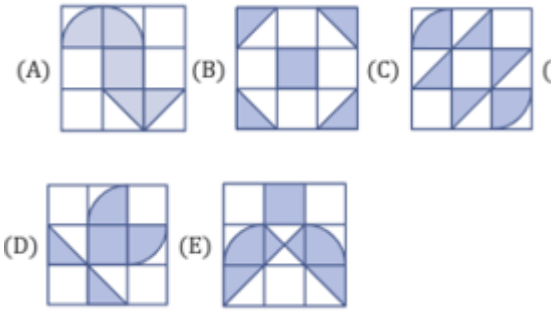
(A) 8
(B) 10
(C) 12
(D) 15
(E) 18

7. Simon thought of a certain number. Then, he made the number twice as big. He then doubled the result, and then doubled it again, and then doubled it once more. Which of the numbers below cannot be the result of this operation?

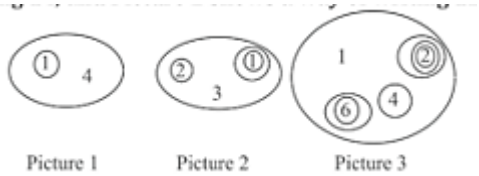
(A) 80
(B) 1200
(C) 48
(D) 84
(E) 880

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8. Which of the shaded regions has the largest area?

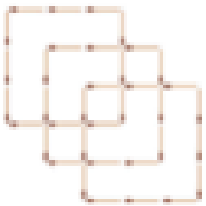


9. Picture 1 shows a way of writing 14, and Picture 2 shows a way of writing 123. What number does Picture 3 represent?



(A) 1246
(B) 2461
(C) 2641
(D) 1462
(E) other answer

10. What is the smallest number of matches you have to add to the puzzle shown in the picture in order to have 11 squares?



(A) 2
(B) 3
(C) 4
(D) 5
(E) 6

4 Points Each

11. Peter runs a lap around the track in 3 minutes and Paul in 4 minutes. Peter and Paul start to run around the track at the same time. After how many minutes will the boys meet at the starting point again?

(A) 6
(B) 8
(C) 10
(D) 12
(E) It depends on the distance around the track.

12. Anya has 201 coins. One third of her coins are 1 dollar coins. The same number of her coins are 2 dollar coins. The rest are 5 dollar coins. How much money does Anya have?

(A) 536 dollars
(B) 201 dollars
(C) 516 dollars
(D) 1020 dollars
(E) 2001 dollars

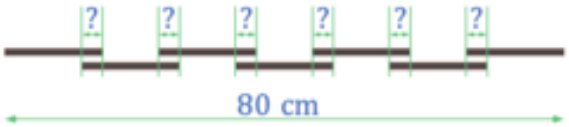
13. At the Olympics there was a 10 kilometer race. Adam did not finish the race and asked his friends the following question: "How many meters did I have left to finish the race if I already ran 8,631 meters, 3,456 decimeters, and 12,340 centimeters?"

(Note: 1 kilometer is 1,000 meters, 1 meter is 10 decimeters, and 1 meter is 100 centimeters.) The answer to his question is:

(A) 1060
(B) 160
(C) 900
(D) 100
(E) 1000

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14. There are 7 sticks, each 14 centimeters long. They were put together in the way shown in the picture, for a total length of 80 centimeters. Each of the segments marked with a question mark is the same length. How long are these segments?



(A) 1 cm
(B) 2 cm
(C) 3 cm
(D) 5 cm
(E) 8 cm

15. If the red dragon had 6 heads more than the green dragon, then both dragons would together have 34 heads. Actually, the red dragon has 6 heads fewer than the green dragon. How many heads does the red dragon have?

(A) 6
(B) 8
(C) 12
(D) 14
(E) 16

16. The length of a certain rectangle is 80 cm and its area is 3,200 cm^2. Find the length of another rectangle if its area and width are half the area and the width of the rectangle described above.

(A) 20 cm
(B) 40 cm
(C) 60 cm
(D) 80 cm
(E) 100 cm

17. Zosia spends 1 hour doing her homework. She spends one third of this time doing math, and she spends two fifths of the remaining time working on geography. How many minutes does she spend doing the rest of her homework?

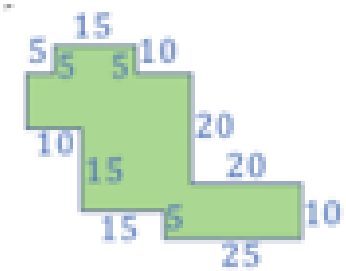
(A) 12
(B) 20
(C) 24
(D) 36
(E) 40

18. The triplets Adam, Ian, and Stan and their sister Mary, who is four years older than they are, altogether were 24 years old three years ago. How old is Mary today?

(A) 5
(B) 8
(C) 9
(D) 12
(E) 15

19. Jonah's garden is shaped in the way shown in the picture. The lengths of its sides are given in meters, and any two adjacent sides are perpendicular. What is the area of the garden in square meters?

(A) 700
(B) 750
(C) 800
(D) 850
(E) 900



20. Together, Adam, Bart, and Charlie earned 280 dollars during their vacation. Adam made twice as much money as Bart and four times as much as Charlie. How many dollars did Charlie earn?

(A) 30
(B) 40
(C) 50
(D) 60
(E) 70

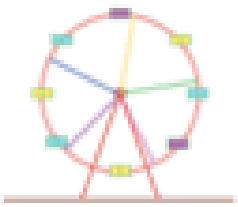
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5 Points Each

21. 9 is the ones digit in the product of 7 x 7. 3 is the ones digit in the product of 7 × 7 × 7. What is the ones digit in the number which is the product of 100 sevens?

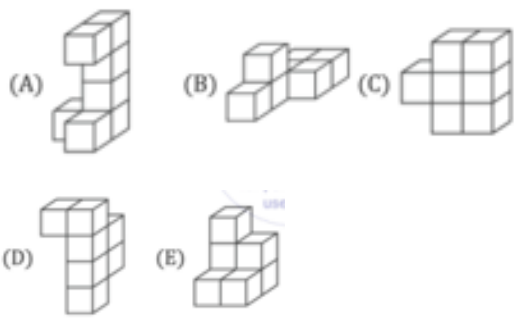
(A) 1
(B) 3
(C) 5
(D) 7
(E) 9

22. The biggest attraction of the Fair is the Big Wheel. The picture shows an example of such a ride but with fewer cabins. The cabins are equally spaced and are numbered consecutively with numbers 1, 2, 3, and so on. When the cabin with the number 25 is at the very top, the cabin with the number 8 is at the very bottom. How many cabins does the Big Wheel have?

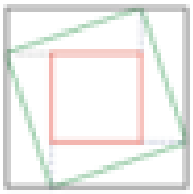


(A) 33
(B) 34
(C) 35
(D) 36
(E) 37

23. Each of the solids below is made of the same number of identical little cubes. For which of these solids would you need the largest amount of paint to paint its entire surface?

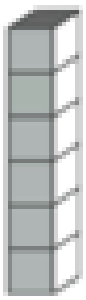


24. The area of the biggest square shown in the picture to the right is 16 cm² and the area of the smallest square is 4 cm². What is the area of the medium square?



(A) 8 cm²
(B) 8½ cm²
(C) 10 cm²
(D) 10½ cm²
(E) 12 cm²

25. The sum of the number of dots on any two opposite sides of any die is 7. Jo is building a solid using six dice which she is gluing together in the way shown in the picture. What is the largest possible number of dots that she can get on the surface of the solid?



(A) 106
(B) 91
(C) 95
(D) 84
(E) 96

26. Fill in the blanks with digits so that the equality is true: 45 x _3 = 3_ _ _. The sum of the four digits placed in the blanks is:

(A) equal to 20
(B) equal to 21
(C) equal to 17
(D) more than 21
(E) less than 17

27. Inside the big cube, which was made out of small cubes, tunnels were made going through the cube in such a way that they are parallel to its walls (see the picture). After making the tunnels, how many small cubes are left in the solid?



(A) 88
(B) 80
(C) 70
(D) 96
(E) 85

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28. A one-hundred-year-old oak generates 1.7 kg of oxygen per hour. How many of these trees do we need to supply oxygen to 34 students for 1 hour if we know that each student consumes 0.7 kg of oxygen per hour by breathing?

(A) 10
(B) 12
(C) 14
(D) 15
(E) 21

29. The vertices of the star shown in the picture are the midpoints of the sides of a regular hexagon. If the area of the star is 6 then the area of the hexagon is:



(A) 8
(B) 9
(C) 12
(D) 15
(E) 18

30. Using digits from 1 to 6, each digit only once, we form two 3-digit numbers, for instance, 645 and 321, and then we subtract the smaller number from the larger one. In this example the difference is 324. What is the smallest possible value of the difference you can get this way?

(A) 69
(B) 56
(C) 111
(D) 47
(E) 38

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Answers key:

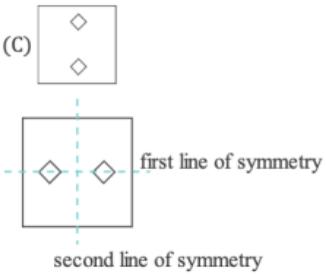
1. B
2. D
3. C
4. B
5. E
6. C
7. D
8. E
9. C
10. A
11. D
12. A
13. C
14. C
15. B
16. D
17. C
18. D
19. E
20. B
21. A
22. B
23. A
24. C
25. E
26. D
27. A
28. C
29. C
30. D

1. (B) 0
Do multiplication before addition, so $2 \times 0 + 0 \times 1 = 0 + 0 = 0$.

2. (D) $\frac{1}{12}$
One box represents $\frac{1}{6}$ of the rectangle, so the shaded area represents $\frac{1}{2} \times \frac{1}{6} = \frac{1}{12}$ of the rectangle.



3. (C)



The top edge of the folded napkin is one line of symmetry. After the first unfolding, the edges on the left become one edge which is also a line of symmetry. In terms of the first line of symmetry, after the first unfolding the paper cut-out looks like a diamond. Then, after unfolding along the second line of symmetry, we see two diamonds. After rotating the unfolded napkin by 90° we get the napkin shown in (C).

4. (B) 8 minutes
24×20 seconds $= 480$ seconds $= 8 \times 60$ seconds $= 8$ minutes, so after 24 hours the clock will be late 8 minutes.

5. (E) 72
The proportion of occupied seats to the empty seats is 2 : 1, so the proportion of occupied seats to all seats is 2 : (2 + 1). The number of passengers on the plane was $\frac{2}{2+1} \times 108 = \frac{2}{3} \times 108 = 2 \times 36 = 72$

6. (C) 12
Ella has (3 – 1) sisters (excluding herself) and (5 + 1) brothers (including Johnny).
The product is $2 \times 6 = 12$.

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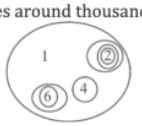
7. (D) 84
Simon doubled his original number 4 times, so the result of this operation is $2 \times 2 \times 2 \times 2 \times$ the original number. The result is a multiple of 16 since $2 \times 2 \times 2 \times 2 = 16$.
On the list only 84 is not a multiple of 16, so it cannot be the result of Simon's operation.
$80 = 5 \times 16$, $1200 = 75 \times 16$, $48 = 3 \times 16$, and $880 = 55 \times 16$.

8. (E) 
Notice that two shaded triangles cover exactly one shaded square, and a shaded circular quadrant (a quarter of a circle) covers more than one shaded triangle.
Let s be the area of one small square. Let c be the area of one circular quadrant.
Then $\frac{1}{2}s < c < s$ (or $s < 2c < 2s$). For each of the regions below we will compute the area in terms of s and c.

(A)  (B)  (C)  (D)  (E) 

For (A) the area is $\left(1+\frac{1}{2}+\frac{1}{2}\right)s + (1+1)c = 2s + 2c$.
For (B) the area is $\left(\frac{1}{2}+\frac{1}{2}+1+\frac{1}{2}+\frac{1}{2}\right)s = 3s$.
For (C) the area is $\left(\frac{1}{2}+\frac{1}{2}+\frac{1}{2}+\frac{1}{2}\right)s + (1+1)c = 2s + 2c$.
For (D) the area is $\left(\frac{1}{2}+1+\frac{1}{2}\right)s + (1+1)c = 2s + 2c$.
For (E) the area is $\left(1+\frac{1}{4}+\frac{1}{4}+\frac{1}{2}+\frac{1}{2}\right)s + (1+1)c = 2\frac{1}{2}s + 2c$.
Notice that $3s = 2s + s < 2s + 2c$ and $2s + 2c < 2\frac{1}{2}s + 2c$, so (B) has the smallest shaded area and (E) has the largest shaded area.

9. (C) 2641
According to Picture 1 and Picture 2, there is no circle around ones digit, there is one circle around tens digit, and two circles around hundreds digit. Following the pattern, we will see 3 circles around thousands digit.

Thus,  represents the number 2641.

10. (A) 2
There are 8 squares visible in the original picture, 3 are large (3×3), 2 are medium (2×2), and 3 are small (1×1). By adding the two matches shown here in pink, three more small squares will be added for a total of 11 squares altogether.



11. (D) 12
The least common multiple of 3 and 4 is 12, so the boys will meet again after 12 minutes.

12. (A) 536 dollars
$201 \div 3 = 67$, so Anya has sixty seven 1 dollar coins, sixty seven 2 dollar coins, and sixty seven 5 dollar coins. Altogether, she has $67 \times (1 + 2 + 5)$ dollars, which is 536 dollars.

13. (C) 900
8,631 meters + 3,456 decimeters + 12,340 centimeters is
8,631 meters + 345.6 meters + 123.40 meters = 9,100 meters.
10 kilometers − 9,100 meters is 10,000 meters − 9,100 meters = 900 meters, so Adam had 900 meters left to finish the race.

14. (C) 3 cm
The seven sticks, each of the length 14 centimeters, cover the length of 80 centimeters. There are 6 overlaps (indicated by "?") of equal length, so the length of one overlap is $(7 \times 14 \text{ cm} - 80 \text{ cm}) \div 6 = 18 \text{ cm} \div 6 = 3 \text{ cm}$.

15. (B) 8
The green dragon and any other dragon with 6 more heads together have 34 heads, so 2 identical green dragons would have $34 - 6 = 28$ heads. Hence, the green dragon has 14 heads. The red dragon has $(14 - 6)$ heads, which is 8 heads.

16. (D) 80 cm
A given rectangle has the length ℓ, the width w, and the area A. Another rectangle has the length ℓ', the width $\frac{w}{2}$, and the area $\frac{A}{2}$, so $\ell \times w = A$ and $\ell' \times \frac{w}{2} = \frac{A}{2}$.
The last equation is equivalent to $\ell' \times w = A$. $\ell \times w = A$, so $\ell' = \ell$. The original rectangle has a length of 80 cm, so the new rectangle also has a length of 80 cm.
Or: The area of a rectangle is found by multiplying the length by the width. If the width is reduced by one-half and the length stays the same, then the area is reduced by one-half. So, the second rectangle has the same length as the first rectangle, that is, 80 cm.

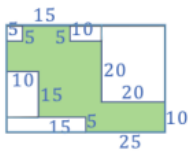
17. (C) 24
1 hour is 60 minutes, so Zosia spends 20 minutes doing math. She spends two fifths of the remaining 40 minutes working on geography, so she works $\frac{2}{5} \times 40$ minutes = 16 minutes on geography. 20 minutes + 16 minutes = 36 minutes and 60 − 36 = 24, so she spends 24 minutes doing the rest of her homework.

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18. (D) 12

If you add 3 years + 3 years + 3 years + 3 years to 24 years, then you see that today the four siblings are altogether 36 years old. To compensate for the age difference between Mary and the triplets add 4 years + 4 years + 4 years to 36 years to see Mary's age being quadrupled. Hence, 4 × Mary's age = 48 years. 48 ÷ 4 = 12, so Mary is 12 years old today.

19. (E) 900



Extend the garden to a full rectangle by adding rectangular parcels as shown in the picture. The top edge of the extended rectangle has length 5 + 15 + 10 + 20 = 50. The right edge of the extended rectangle has length 5 + 20 + 10 = 35, so the area of the extended rectangle is 35 × 50 = 1,750.
The areas of the added rectangular parcels (counted along the top edge and then along the left edge) are: 5 × 5, 5 × 10, (5 + 20) × 20, 15 × 10, and 5 × (10 + 15). Their sum is 25 + 50 + 500 + 150 + 125 = 850.
1,750 – 850 = 900, so the area of Jonah's garden is 900 square meters.

20. (B) 40

Adam's money is 4 times Charlie's money and Bart's money is 2 times Charlie's money, so the three of them together earned 4 + 2 + 1 times Charlie's money. Therefore, Charlie earned 280 ÷ 7 = 40 dollars.

21. (A) 1

7 × 7 × 7 ends with the digit 3, so 7 × 7 × 7 × 7 = (7 × 7 × 7) × 7 ends with the last digit of 3 × 7, which is 1. $\underbrace{7\times7\times7\times7\times7\times7\times7\times7\times\ldots\times7\times7\times7\times7}_{100\ times}$ is $\underbrace{(7\times7\times7\times7)\times(7\times7\times7\times7)\times\ldots\times(7\times7\times7\times7)}_{25\ times}$, so its last digit is the last digit of $\underbrace{1\times1\times\ldots\times1}_{25\ times}$, which is 1.

22. (B) 34



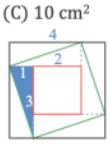
Moving clockwise from the cabin with number 8 to the cabin with number 25, we encounter cabins with the following numbers: 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, and 24. There are 16 of them. By symmetry, there are also 16 cabins when we move counterclockwise from 8 to 25. Together, not counting the cabins with numbers 8 and 25, there are 32 cabins. Including those two cabins, the Big Wheel has 34 cabins.

23. (A)



Two cubes are adjacent if they share a face. For a given cube, the number of its faces to be painted depends on the number of cubes adjacent to the given cube. So, for a given cube the number of faces to be painted is 6 minus the number of adjacent cubes. For each solid we count (from top to bottom, left to right, front to back) the number of faces to be painted.
(A) 5 + 4 + 4 + 4 + 5 + 3 + 5 = 30, (B) 5 + 5 + 3 + 3 + 4 + 4 + 4 = 28,
(C) 4 + 4 + 5 + 2 + 3 + 4 + 4 = 26, (D) 5 + 4 + 3 + 4 + 3 + 4 + 5 = 28, and
(E) 5 + 3 + 4 + 3 + 4 + (invisible) 3 + 4 = 26.
Thus, the solid (A) requires the largest amount of paint.

24. (C) 10 cm²



Each side of the biggest square is 4 cm, since 4 cm × 4 cm = 16 cm². Each side of the smallest square is 2 cm, since 2 cm × 2 cm = 4 cm². So, the legs of the shaded right triangle are 1 cm and 3 cm. The medium square consists of four such triangles and the smallest square, so the area of the medium square is 4 × (½ × 3 × 1) cm² + 4 cm², which is 10 cm².

25. (E) 96

Each die, without counting the top and bottom faces of the solid, contributes two pairs of opposite faces to the surface of the solid. The numbers of dots on each pair of opposite faces add up to 7; so, each die (without counting the top and bottom faces of the solid) contributes 14 dots to the surface of the solid. Without counting the top and bottom faces of the solid, the six dice together contribute 6 × 14 = 84 dots to the surface of the solid.
To have the largest number of dots on the surface we need 6 dots on the top face and 6 dots on the bottom face, so the largest number of dots on the whole surface is 84 + 6 + 6, which is 96.

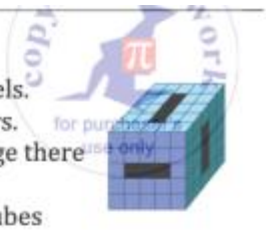
26. (D) more than 21

The number 3_ _ _ is between 3,000 and 3,999. 3,000 ÷ 45 = 200 ÷ 3 = $66+\frac{2}{3}$ and 3,999 ÷ 45 = 1,333 ÷ 15 = $88+\frac{13}{15}$, so 67 ≤ _3 ≤ 88 since _3 is a natural number and 45 × _3 = 3_ _ _. Natural numbers between 67 and 88 with the last digit 3 are 73 and 83.
In the first case, 45 × 73 = 3285. The sum of the digits in the blanks is 7 + 2 + 8 + 5, which is 22. In the second case, 45 × 83 = 3735. The sum of the digits in the blanks is 8 + 7 + 3 + 5, which is 23. In either case, the sum is not 20, it is not 21, it is not 17, and it is not less than 17, but it is more than 21.

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27. (A) 88



The small cubes along the edges of the big cube are not affected by tunnels. Among these cubes there are 8 corner cubes since any cube has 8 corners. The other small cubes along the edges do not overlap and along each edge there are $(5-2)=3$ such cubes. A cube always has $3\times 4=12$ edges (three directions and 4 edges in each direction), so the total number of small cubes along the edges is $8+12\times(5-2)=44$.

In the top layer of the big cube there are still $2\times 3=6$ small cubes (along the tunnel opening) not affected by the tunnels and not counted yet. The same applies to any of the 6 outer layers of the big cube and these small cubes do not overlap. Therefore, the number of all small cubes of the six outer layers which are not affected by the tunnels is $44+6\times(2\times 3)=44+36=$ $=80$. Take away these 80 cubes from the big cube. You are left with a $3\times 3\times 3$ cube and three mutually perpendicular tunnels through the center of the $3\times 3\times 3$ cube. Only the 8 corners of the $3\times 3\times 3$ cube are not affected by the tunnels, so the total number of small cubes left in the solid is $80+8=88$.

28. (C) 14

In one hour the students consume (34×0.7) kg of oxygen, so the number of oaks we need is (34×0.7) kg $\div$ 1.7 kg. $\frac{34\times 0.7}{1.7}=\frac{34\times 7}{17}=14$, so 14 such oak trees are needed.

29. (C) 12



Draw 3 line segments through the center of the hexagon, parallel to the sides of the hexagon, as shown in the picture. The whole hexagon is divided into 12 congruent rhombuses, 6 of them shaded and 6 of them white. Hence the shaded region of the hexagon (the star) is half the area of the hexagon. Therefore, the area of the hexagon is $2\times 6=12$.

30. (D) 47

Let *abc* and *def* be 3-digit numbers such that *a*, *b*, *c*, *d*, *e*, *f* (in some order) are the digits 1, 2, 3, 4, 5, and 6, and $a<d$. The difference is $100\times(d-a)+10\times(e-b)+(f-c)$.

Intuitively, keep a and d as close as possible with $a<d$, keep b and e as far apart as possible with $e<b$, and keep f and c also distant with $f<c$ but respect the other requirements.

Thus, a and d should be selected form the middle of the list 1, 2, 3, 4, 5, 6 and e and b from the start and the end of the list.

365 and 412 are such numbers and the difference is $412-365=47$.

47 is the actual answer.

Indeed, if $(d-a)\geq 2$, then the difference is greater than 100.

If $(d-a)=1$ but $(e-b)$ is not -5, then the difference is greater than 50.

Thus, we need $(d-a)=1$ and $(e-b)=-5$, so $e=1$ and $b=6$ are necessary for the smallest difference. After that $f=2$ and $c=5$ is the optimal option left, so a must be 3 and d must be 4.

Clearly, $412-365=47$ is the smallest possible difference.

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